Giovanni Tauro

R&D Engineer, PhD Candidate| Human Behaviour & Data Analysis



Bari-Lecco, Italy +39 3279437161 giovanni.taurodi@gmail.com
LinkedIn GitHub Scholar Portfolio GrabCAD

R&D Engineer and PhD Candidate with 5+ years of applied research at CNR-STIIMA on how people behave and respond to feedback. With OmniTrack I ran navigation experiments with 16 blind participants, collecting position and IMU data and analysing in Python how they respond to multimodal feedback. Research stays in experimental psychology (Utrecht University) and behaviour analysis (Toyo University).

Education

2023–27 **PhD, Mechanics and Advanced Engineering Sciences** *Bologna University, IT*

PhD Candidate, expected 2027. Topic: IoT-based Multimodal Navigation System – OmniTrack
- Research Mobility, Utrecht University, NLD: Sensory navigation and experimental psychology.
- Research Mobility, Toyo University, JP: Feedback algorithm and behaviour analysis.

2017–20 **MSc, Design & Engineering** *Politecnico di Milano, IT*

Topic: UX development of an ultrasonic dental device: from concept to validation: WashB
- Research Mobility, Chiba University, JP: UX research studies for Fujitsu and Panasonic.

2012–15 **BSc, Industrial Design** *Florence University, IT*

Topics: Foundation in research methodologies, development, and critical thinking.

Work Experience

R&D Product Engineer, STIIMA, Lecco
Feb 2021 – Present

- Developed sensor-based rehabilitation and assistive technology systems, translating clinical requirements and user needs into functional prototypes for experimental validation .
- Defined the architecture of multimodal IoT feedback platforms, integrating wearable sensing, indoor localization, BLE communication, and adaptive audio, visual, and haptic feedback.
- Delivered 3+ functional prototypes across embedded electronics, wearable devices, and interactive systems, supporting projects from concept development to ISO compliance technology delivery.
- Designed communication infrastructures and evaluation protocols for digital health applications, assessing usability, system performance, and user interaction through experimental testing.
- **Head of Product (Co-Founder)**, Rehabilia Technologies s.r.l. (May 2023 – Present), applying this R&D expertise to lead product development as Head of Product:
  - Led end-to-end product development of rehabilitation technologies, from user research and problem discovery to validation and product delivery.
  - Defined product requirements and priorities by translating user needs, business goals, and stakeholder feedback into actionable roadmaps.
  - Collaborated with engineering, design, and clinical stakeholders to align product strategy and deliver iterative product improvements.
  - Validated product decisions through usability testing, interviews, and data-driven loops.

Embedded Systems • IoT • ESP32 • Arduino • PlatformIO • VS • BLE GATT • ROS2 • C++ • C# • Python • Prototyping • 3DCAD • FDM • Material Selection • UX Research • Product Development • Stakeholder Communication • ASANA • Cross-functional Teamwork • Project Management • Public Speaker

Freelancer, Florence
Jan 2016 – Sep 2017, collaboration with DIMA3D

- Delivered end-to-end reverse engineering workflows, converting 3D scan data into validated parametric CAD models and production-ready assets.

CAD • Reverse Engineering • Geomagic • SolidWorks • 3D Scanning • Product Development • Technical Documentation • Manufacturing Preparation

Selected Projects

OmniTrack: IoT-based Multimodal Navigation System /Frontiers

GIT_1
GIT_2
GIT_3

- Built end-to-end hardware/software prototypes, validated with 16 blind participants, integrating IoT-oriented sensing, processing, and user feedback into a complete system.
- Applied real-time data processing and system integration principles to transform physical-world signals into actionable information.

Python • Data Anlysis • C++ • JS • Dashboard • ESP32 • ROS2 • BLE GATT • Sensors • I2C • Linux

PhiCube: reconfigurable bilateral robotic device for neurorehabilitation /Frontiers

WEB

- Designed a modular rehabilitation device combining UX research, mechanical design, embedded systems, and multimodal interaction.
- Validated with 19 children (SUS 85.8/100); 134 patients in two on-going multicentric studies.

UX • Interaction Design • CAD • Prototyping • Assembly • Usability

RehaBalloon: Interactive Platform

GIT

- Prototyped a low-cost rehabilitation device in 24 hours during the LMU Munich "Hacking with Hardware" challenge, integrating pressure sensing, visual feedback, and embedded electronics.
- Implemented a TensorFlow Lite model on Arduino embedded machine learning classifier exercise patterns and record rehabilitation activity metrics in real time.

Seeed Studio • Sensors • C++ • TensorFlow • Machine Learning

Certifications

2026 **Claude 101 Badge — Claude Code Badge — Claude** *Anthropic, Online*
2024 **.ppt — Robotics and Allied Technologies for Rehab** *MUR, Rome, Italy*

Patent

2023 Malosio, M., Nicora, M. L., Redaelli, D., Tauro, G. WO2023214349A1 *Device for bilateral rehabilitation*.

Languages

Italian: Native English: Advanced (IELTS 2017) French: Basic (A2)

Lecco, October 9, 2026

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